Enabling DataEngine on a VAST Cluster Tenant

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Article summary

Overview

VAST DataEngine enablement requires prerequisite configurations that are external to the VAST cluster as well as prerequisite configurations on the VAST cluster itself.

Additionally, there are configurations on the VAST cluster that are required for DataEngine usage and are not prerequisites to enablement.

High-Level Steps

To enable VAST DataEngine on a VAST Cluster tenant, follow these steps in this order:

1. Establish prerequisite external services.
2. Configure Prerequisites on the VAST Cluster. These are configurations that you must be established on the VAST Cluster tenant before you enable DataEngine.
3. Enable DataEngine. In this step, you will be prompted to connect the prerequisite external services to the tenant.

Establish Prerequisite External Services

VAST DataEngine connects with the following services that are external to the VAST Cluster. They must be configured before you start:

- One or more container registries. Used to store images of the functions deployed by DataEngine. The container registry must be accessible over https and have a valid CA certificate.
- One or more Kubernetes clusters. Used to execute DataEngine pipelines and functions.
- Optionally, a third-party event broker. DataEngine requires an event broker. This can be either a third-party event broker or the VAST Event Broker. If you choose to deploy DataEngine with the VAST Event Broker, you do not need to configure it before the enablement procedure.

Preparing Container Registries

A container registry is a service for storing and distributing container images. You can use a registry service that supports authentication with user name and password, such as a Docker Hub hosted registry service, or a registry that supports a stable authentication secret, such as a AWS ECR hosted registry. You will need to connect the tenant's DataEngine to at least one container registry. You will store images of the functions you want to deploy in the container registry.

Preparing Kubernetes Clusters

Install the VAST Zarf DataEngine package on each Kubernetes cluster in your environment that you intend to use to deploy VAST DataEngine functions and pipelines. This package installs the following services on the Kubernetes cluster:

- VAST Operator. VAST's Kubernetes operator, which enables deployment of VAST pipelines and functions on the Kubernetes cluster.
- VAST Telemetry Collector. Provides observability logs and traces over pipelines and functions executed on the Kubernetes cluster by DataEngine.
- *Knative* components used by DataEngine.

For environments where Zarf is not already in use for managing Kubernetes deployments, you will also install the Zarf agent and the VAST Zarf mutator from the package.

To install the VAST Zarf DataEngine package on each Kubernetes cluster, do one of the following:

- Installing the VAST Zarf DataEngine package (Zarf Not Already in Use)
- Installing the VAST Zarf DataEngine package (Zarf Already in Use)

Installing the VAST Zarf DataEngine package (Zarf Not Already in Use)

Use this procedure if Zarf is not already actively being used to manage your Kubernetes cluster deployment.

> **Note**
>
> To check if Zarf is actively being used to manage your Kubernetes cluster deployment:
>
> Plain text COPY
>
> ```
> $ kubectl get pods -n zarf
> NAME                                 READY   STATUS    RESTARTS   AGE
> agent-hook-6c97484f98-5q9jj          1/1     Running   0          10m
> ```

```
agent-hook-6c97484f98-tgcjf              1/1     Running   0          10m
zarf-docker-registry-fd8859657-4rlm9     1/1     Running   0          11m
```

If Zarf is in fact already deployed, switch to **Installing the VAST Zarf DataEngine package (Zarf Already in Use)** .

1. Ensure you have a valid *Kubeconfig* file for connecting to the Kubernetes cluster.
2. Verify that the `inotify` limits are large enough to allow for VAST Zarf DataEngine package installation:
   - To check current limits:

     Plain text COPY

     ```
     sysctl fs.inotify.max_user_instances
     sysctl fs.inotify.max_user_watches
     ```

   - To increase the limits temporarily:

     Plain text COPY

     ```
     sudo sysctl fs.inotify.max_user_instances=8192
     sudo sysctl fs.inotify.max_user_watches=524288
     ```

   - To make the change permanent:

     Plain text COPY

     ```
     echo "fs.inotify.max_user_instances=8192" | sudo tee -a /etc/sysctl.co
     echo "fs.inotify.max_user_watches=524288" | sudo tee -a /etc/sysctl.co
     sudo sysctl -p
     ```

3. Obtain the VAST Zarf DataEngine package, `vast_dataengine_release_<release number>_<pipeline id>.tar.gz`.

   Note

   If you are running a version of VAST Cluster prior to 5.4.1-sp2, run this command before you proceed:

   Plain text COPY

   ```
   if [ -f vast-zarf-mutator-amd64-v0.60.0.tar.zst ]; then
     mv vast-zarf-mutator-amd64-v0.60.0.tar.zst zarf-init-amd64-v0.60.0.tar.zst
   fi
   ```

4. Extract the Zarf agent init package from the VAST Zarf DataEngine package (for example `zarf-init-amd64-v0.60.0.tar.zst`) and run `zarf init` with the following options :

Plain text

COPY

```
zarf init --architecture amd64 zarf-init-amd64-v0.60.0.tar.zst --set REGI
```

Note

The above call assumes that a default storage class exists. Otherwise, add the `--storage-class` option to the call. For example `--storage-class=local-path`.

5. Add the following labels to the Kubernetes namespace:

Plain text

COPY

```
create_namespace_for_zarf(){
local namespace=$1
kubectl create ns $namespace || true
kubectl label namespace $namespace zarf.dev/vast=mutate
}
create_namespace_for_zarf vast-dataengine
create_namespace_for_zarf knative-eventing
create_namespace_for_zarf knative-serving
```

6. Extract the VAST Operator file (for example, `zarf-package-dataengine-amd64-1.0.0.tar.zst`) and run `zarf package deploy` with the following options:

Plain text

COPY

```
zarf package deploy --architecture amd64 zarf-package-dataengine*.tar.zst
```

Installing the VAST Zarf DataEngine package (Zarf Already in Use)

Note

This procedure can also be used to update the VAST Zarf DataEngine package.

Use this procedure if Zarf is already in use for managing your Kubernetes cluster deployment.

1. Ensure you have a valid *Kubeconfig* file for connecting to the Kubernetes cluster.
2. Verify that the `inotify` limits are large enough to allow for VAST Zarf DataEngine package installation:
   - To check current limits:

Plain text COPY

```
sysctl fs.inotify.max_user_instances
sysctl fs.inotify.max_user_watches
```

- To increase the limits temporarily:

Plain text COPY

```
sudo sysctl fs.inotify.max_user_instances=8192
sudo sysctl fs.inotify.max_user_watches=524288
```

- To make the change permanent:

Plain text COPY

```
echo "fs.inotify.max_user_instances=8192" | sudo tee -a /etc/sysctl.co
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sudo sysctl -p
```

3. Obtain the VAST Zarf DataEngine package, `vast_dataengine_release_<release number>_<pipeline id>.tar.gz`.
4. Extract the VAST Operator file (for example, `zarf-package-dataengine-amd64-1.0.0.tar.zst`) and run `zarf package deploy` with the following options:

Plain text COPY

```
zarf package deploy --architecture amd64 zarf-package-dataengine-amd64*.t
```

Configure Prerequisites on the VAST Cluster

This procedure uses the VAST Web UI to create all the prerequisite configurations necessary on a given VAST Cluster tenant for enabling DataEngine.

- If you want to enable DataEngine on a new tenant rather than the default tenant or another existing tenant, create a new tenant on the (VAST) cluster.
- If you want to use a third-party event broker as the default event broker for DataEngine, configure the third-party event broker.
- If you want to use VAST Event Broker as the default event broker for DataEngine, Provision a Bucket Owner User

Create a New Tenant (Optional)

If you want to enable DataEngine on a new tenant rather than the default tenant or another existing tenant, create a new tenant on the (VAST) cluster.

To create a new tenant with minimal settings:

1. From the left navigation menu, select **Element Store** and then **Tenants**.
2. Click **Create Tenant** to open the **Create New Tenant** dialog.
3. In the **General** tab, complete the fields:

Tenant name	Enter a name for the tenant.
Domain	The domain name for the tenant. The domain name is used to build the cluster' VMS login URL for the tenant, as shown in the preview below: `https://<VMS IP>/#/login/<domain name>`. If a domain name is not specified, the tenant name is used: `https://<VMS IP>/#/login/<tenant name>`. **Note** The domain name is case-insensitive.

4. Click **Create**. The tenant is created and appears in the list of tenants in the **Tenants** page.

Configure a Third-Party Event Broker

> **Note**
>
> If you are going to use VAST Event Broker as the default event broker for DataEngine, you can skip this step.

1. Go to **Settings** → **Notifications** → **Notification Kafka Brokers**.
2. Click **Add a New Kafka Broker** and complete these fields:

Name	Enter a name for the event broker configurati
Tenant	Select one tenant for which the event broker be available, or leave the default of *All tenant* that the broker is available for all tenants.

Host	Enter the bootstrap URL of the event broker server. You can specify an IP or FQDN. If the Kafka cluster runs multiple event broker click **+Add** to add more hosts. You can add u five hosts.
Port	Enter a port to communicate with the event broker server.

Tip

Ensure that the hosts are accessible from the VAST cluster's management interface at the specified ports.

3. Click **Add Kafka Broker**.

   The newly created event broker configuration is added to the list of event brokers.

Provision a Bucket Owner User

Note

If you are going to configure an event broker external to the cluster as the default event broker for DataEngine, you can skip this step.

If you are going to use a VAST Event Broker (a view with the "Kafka" protocol enabled) as the default event broker that DataEngine will use to stream events, you need to provision at least one user with S3 access keys and permission to create buckets on the tenant. This is because when you are enabling DataEngine, you will create the VAST Event Broker and you will need to reference a user as a bucket owner for the view.

You will also need to reference a user of this type as a bucket owner when you provision S3 bucket views from which triggers can consume events, although this is not an essential step for enabling DataEngine.

In order to be able to grant multiple users to access DataEngine and to create triggers using the same VAST Event Broker, you also need to make sure this user belongs to a group that you will use for all non VMS-manager users that you want to allow to access VAST DataEngine on the tenant, once it's enabled. This is because you will need to give those users bucket listing permission for the VAST Event Broker view (this is done in the next step).

1. Make sure the provider you want to use is attached to the tenant. The provider can be a VAST provider, LDAP, or Active Directory. The provider should be configured on the cluster. For general information about managing providers, see Providers.

   > **Note**
   >
   > If you use a VAST provider, enable the provider for *cluster admin* users and *tenant admin* users to manage and view, as needed. The default VAST provider is initially enabled for management by *cluster admin* users only.

   To attach a provider to a tenant:

   - In the **Tenants** tab of the **Element Store** page, right-click the tenant and select **Edit**.
   - Under **Providers and User Access**, select the provider that you want to connect to the tenant.
   - Click **Update**.

2. Create a user on the provider. Grant the user *bucket create* and *bucket delete* permissions.

   > **Tip**
   >
   > It is usually best practice that the user belongs to a group designated for granting DataEngine access. You will be able to grant the user's group bucket listing permission for a VAST Event Broker and you will be able to grant the same group **access to DataEngine**.

   > **Tip**
   >
   > To create a user as part of a group on a VAST provider:
   >
   > 1. From the left navigation menu, select **User Management** and then, under **Users and Groups**, **Local Groups**.
   > 2. On the **Groups** page that opens, click **+ Create Group**.
   > 3. In the **Add Group** dialog, complete the fields:
   >
   > | Field | Description |
   > |---|---|
   > | **Name** (required) | Enter a name for the group. |
   > | **GID** (required) | Enter a POSIX group ID (GID) for the |
   > | **Local Provider** (required) | Select the VAST provider with which be associated. |
   >
   > 4. Click **Create**. The group is created.
   > 5. Switch to the **Local Users** tab.

6. Click **Create User** and complete the following fields:

Field	Description
Name (required)	The user name.
UID	The user's POSIX UID.
Local Provider (required)	Select the VAST provider with which be associated.
Leading group	The name of the user's leading group This is the group assigned by default owning group of any files created by Select the group you created in the p from the dropdown.
Groups	Names of other groups that the user beside the leading group. Also known groups. Select groups from the dropdown. If not been added to the VAST provide **group** first.
Select tenant to see user details	Select the tenant from the list. Tenan with the selected VAST provider (if ar shown, as well as the default tenant.
Allow Create Bucket	Enable this setting. (Alternatively, you this permission through an identity p attached to either the group or the u
Allow Delete Bucket	Enable this setting. (Alternatively, you this permission through an identity p attached to either the group or the u

3. Do one of the following, as appropriate, to grant the user an S3 access key pair:
   - Edit the user you just created (if its a local user) and choose to generate an access key pair.
   - Query the user (this is applicable to a user on any provider) and generate an access key pair for the user:
     1. In the **Query User or Group** tab of the **User Management** page, query the user on the relevant provider.
     2. In the query result pane on the right, under **Access Keys**, click **Add New Key**.

Enable DataEngine

This procedure enables DataEngine on a tenant. The procedure includes selecting or creating an event broker and event broker topics, connecting the tenant to a container registry and connecting the tenant to a k8s cluster.

Enabling Data Engine from the VAST Web UI

1. From the left navigation menu, select **Element Store** and then **Tenants**.
2. Right-click the tenant and select **Enable DataEngine**.

   The **Enable DataEngine** dialog opens.
3. In the **Assign Kafka** step screen, from the **Default Broker** dropdown, do one of the following:
   - To use a third-party event broker, select the event broker from the dropdown.
   - To use a VAST Event Broker, select **Add new broker**.

     > **Note**
     >
     > If the VAST Event Broker is already created, you can select it from the dropdown.

     The **Add View** dialog opens. Some of the fields are pre-filled. Continue with the following steps to configure a VAST Event Broker:

     1. Complete the following fields:

Path	Specify the path in the tenant's Elem where the Event Broker should resid must begin with a slash.
Create directory	Enable this.
S3 bucket name	Enter a name for the S3 bucket in wh Event Broker will reside.
Bucket Owner	Provide the user you created in Prov Owner User. Click to enter the name typing and then select a name from
Authentication Methods	Set these as you require. For informa these settings, see Configuring User Authentication for the VAST Event B

     2. Create or select a suitable view policy. The view policy needs to have S3 Native security flavor enabled. The policy should also grant S3 bucket listing permission to the user group that you are using to provision application users, although you could add the bucket listing

permission at a later stage. Bucket listing permission will enable all group members to create triggers using the event broker you are creating.

- To use an existing policy, select the policy from the **Policy name** dropdown.
- To create a new view policy:
  1. In the **Policy name** dropdown, select **Add new Policy**.
  2. In the **Add Policy** dialog, fill the following fields:

Tenant	Select the relevant tenant.
Name	Enter a name for the policy.
Security Flavor	Select **S3 Native**.
Group membership source	Select any of the options. T for view policies, although a this use case.

  3. In the **S3** section, in the **Bucket listing permission (groups)** field, enter the name of the user group that you are going to use to provision users' access to DataEngine.

     > **Note**
     >
     > This step is not strictly mandatory for enabling DataEngine. It is typically needed, assuming you are using a VAST Event Broker and granting multiple users access to DataEngine. Giving all users bucket listing permission in this policy will enable the users to create triggers using a VAST Event Broker created with a group member as the bucket owner.

  4. Click **Create**.

     The policy is created and is selected for you in the **Policy name** field ofthe **Add View** dialog.

3. Create or select a virtual IP pool for the event broker. The virtual IP pool needs to include at least as many virtual IP addresses as there are CNodes on the cluster. It also needs to be dedicated exclusively to the VAST Event Broker.
   - To use an existing virtual IP pool, select the virtual IP pool from the **VIP Pool** dropdown.
   - To create a new virtual IP pool:
     1. In the **Kafka** section, from the **VIP Pool** dropdown, select **Add new VIP pool**.
     2. In the **Add Virtual IP Pool** dialog, fill the following fields:

VAST Web UI Field Name	Requirement
Tenant	Specify the tenant on whicl enable DataEngine.
Name	Give this a suitable name fc IP pool. This virtual IP pool exclusively for the VAST Ev
Role	Set this to **Protocols**.
Subnet CIDR IPv4/Subnet CIDR IPv6	Set this as needed for the I you supply.
IP Range List	Supply a range of IP addres many IPs as there are CNoc pool of IP addresses will be to the VAST Event Broker fc

3. Click **Create**.

   The virtual IP pool is created and is selected for you in the **VIP Pool** field ofthe **Add View** dialog.

4. Click **Create**.

4. From the **Default Topic** dropdown, select **Add new topic** to create the default topic on the event broker. Triggers will be streamed to this topic by default and it will be available immediately for hosting triggers when DataEngine is enabled. In most cases, there is no need to create additional topics for hosting triggers. Assuming the default topic will be used for all or most triggers, it's advisable to set wide partitioning, probably in the region of 50-100 partitions.
5. In the **Create Topic** dialog, complete the following required fields:

Topic Name	Enter a name for the topic.
Number of partitions	Enter the number of partitions for the topic. Each partition can hold up to 1000 topics. Th number of partitions in a topic cannot be cha after the topic has been created. The default topic will be available immediatel hosting triggers when DataEngine is enabled. Unless your use case specifically requires mu topics, the default topic may be used for hos all triggers. Therefore, it is usually advisable tc

	wide partitioning, probably in the region of 50 partitions.
Retention period	Specify the amount of time to keep an event record in the topic. When the retention period a record expires, the record is deleted from th topic. The default retention period is seven da The minimum allowed retention period is 6 ho
Compaction	If this flag is set, VAST Cluster keeps only the latest record version for each key in the parti log. Any previous versions of the record are deleted. **Note** Compaction is performed asynchronously i the background. At some point in time, duplicate keys may exist. Select this checkbox to enable compaction in topic. With compaction, the topic preserves the latest version of each message key in eac partition.
Advanced Options	
Timestamp source	Select the source to use for the event time st • **Time event is created (producer)**. Sets t event timestamp based on the time when event was encountered at the event producer. • **Time event is logged (server)**. Sets the e timestamp based on the time when the e record was added to the log at the event broker.
Event time evaluation period	The maximum acceptable time between the e time and server time. • **Infinity**. For **Time event is created (producer)** timestamps, this option speci

	that any difference between the produce time and the server time is acceptable, e the message timestamp can be earlier or than the broker timestamp for any amou time. • **Define Manually**. For **Time event is crea (producer)** timestamps, specifies the acceptable difference between the prod time and the server time: ◦ **Not before**: Determines how much earlier the message timestamp can b than the broker timestamp. If this va exceeded, the message is rejected. ◦ **Not after**: Determines how much lat the message timestamp can be than broker timestamp. If this value is exceeded, the message is rejected.

6. Click **Create**.
7. In the **Default Deadletter Topic** dropdown, select **Add new topic** to create the deadletter topic, which is the topic to which events are routed in case of function failures, routing failures and dispatching failures. The deadletter topic will not be available to configure as the topic for any trigger event.
8. Click **Add Certificate** to upload a TLS certificate.
9. Paste a CA certificate into the box provided and click **Upload Certificate**.
10. Click **Next**.
11. In the **Link Kubernetes Cluster** step, complete these fields:

Name (required)	Enter a name for the Kubernetes cluster to wl you want DataEngine to connect.
API Server URL (required)	Enter the Kubernetes API server endpoint UR For example: *https://kube-server-57:6443*
Description	Enter a description of the Kubernetes cluster. is optional.

12. In the **Certificates** area, use the fields provided to upload the CA certificate, client certificate and private key used for authenticating over mTLS to the Kubernetes cluster. For each certificate, click **+Add new** , paste the certificate content, base64-encoded, and then click **Save Client certificate /Save CA certificate/Save Private Key**.

13. In the **Tags** area, enter any metadata tag that you want to tag the Kubernetes cluster resource with, and select **Add Tag** to add the tag. Repeat for additional tags. This is optional.
14. Click **Next**.
15. In the **Select Namespaces** step, the namespaces on the Kubernetes cluster are listed on the left under **All possible Properties**. Use the selection boxes and arrow buttons to choose which of the namespaces to allow the tenant's DataEngine to access and list them on the right under **Selected Properties**. Pipelines will be deployed in the selected namespace(s).
16. Click **Next**.
17. In the **Link Container Registry** step, complete the fields to connect the container registry:

Primary kubernetes cluster	Select the primary Kubernetes cluster from th dropdown.
Additional kubernetes clusters	Select any additional Kubernetes clusters tha configured on the tenant if you want them to able to deploy any function that has its image stored on the container registry.
Name	Enter a name for the container registry.
Base URL	Enter the base URL of the container registry.
Description	Enter a description for the container registry. (Optional.)
Authentication Method	Select the authentication method required by container registry for the DataEngine to authenticate to the container registry: • **User credentials**. Choose this option if a name and password is required. In the **Username** and **Password** fields, enter va user credentials for authenticating to the container registry. • **Kubernetes secret**. Choose this option i stable authentication secret is required. I **Kubernetes Secret Name** field, enter th name of an existing Kubernetes Secret th contains the credentials for authenticatir the container registry . • **None**. Choose this option if no authentic is required to connect to the container

	registry.
Tags	Optionally, you can use this section to tag th container registry with any meaningful text st for ease of organization and identification. To add a tag, enter a string in the **Tag** field an then click **Add Tag**. The tag is added to the **Tags** list. If you want to remove a tag that you added, c the tag's removal icon.

18. Click **Finish**.